Support-Conditioned Evaluation Auditing for Aerial Vehicle Detection

Xinling Wen, Jiahao Zhang, Wenqi Liu, Yangxu Ren, and Yu Chen

***Abstract*—Aerial-detection scores can change because a small-object rule changes the annotation support being scored. We formulate support-conditioned evaluation as one chain: annotation-support mismatch, controlled support perturbation, metric uncertainty, and rank robustness on common coverage. The support layer reports retained valid ground truth under absolute and normalized rules. The perturbation layer separates target-set removal from source-ignore treatment and reports a policy band. The rank layer compares frozen configuration pools only on their shared image set and pairs margins with image-paired bootstrap intervals. At 24 pixels, the retained vehicle support is 73.3%, 95.1%, and 9.7% for VisDrone, local UAVDT, and AI-TOD; at normalized support .015 it is 89.8%, 97.4%, and 81.0%. In a five-configuration, 548-image VisDrone pool, headline Y11m–ASD F_1 margins are .0449 [.0391, .0506] at 24 pixels and .0277 [.0234, .0321] at .015, whereas observed rank-boundary cells have intervals spanning zero. The resulting audit distinguishes a changed evaluation target, a policy-sensitive metric, and an uncertainty-qualified within-pool comparison.**

***Index Terms*—Aerial detection, annotation support, evaluation sensitivity, rank robustness, vehicle detection.**

I. Introduction

A reported detection score is meaningful only after its scored target is fixed. In aerial imagery, a support rule can change that target sharply: the same pixel cutoff may retain most vehicles in one source while removing most vehicles in another. Resolution, ontology, source-provided ignores, and tiny-object conventions therefore enter an evaluation result before one compares detector outputs. This issue is consequential for heterogeneous aerial benchmarks [1], [2], [3], [4] and complements research on dataset bias, label quality, and error diagnosis [5], [6], [7].

Recent GRSL studies illustrate both the importance of high-resolution remote-sensing processing and the sensitivity of validation to geolocation conditions [8], [9]. What remains underreported in cross-artifact detection comparisons is a connected account of *what annotation support changed*, *how the metric reacts to that change*, and *whether the resulting model conclusion holds on a common denominator*. Reporting many evaluation switches without this ordering obscures the scientific question.

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Several adjacent lines motivate this account. Unified label spaces address semantic compatibility across datasets [10]; slicing and other tiny-object procedures change the prediction process itself [11]; noise-aware evaluation and label-error screening study annotation reliability [12], [13], [14]; and error taxonomies localize detector failure modes [7]. Our focus begins after an artifact and prediction cache are fixed: it records how an explicit change in scored support propagates into a metric and then into a within-pool decision.

We introduce a support-conditioned audit with the following three layers.

1) *Annotation-support mismatch:* report raw prediction coverage and retained valid ground truth under absolute and normalized support rules.
2) *Controlled support perturbation:* hold prediction artifacts fixed, vary the target/ignore contract explicitly, and summarize the resulting metric uncertainty by a policy band.
3) *Rank robustness on common coverage:* compare a frozen configuration pool only on shared images, map finite-grid winner regions, and pair a point margin with image-paired uncertainty.

Together, the three layers turn a reported score into a traceable support-to-decision record.

II. Support-Conditioned Audit

A. Artifacts, mapping, and coverage

We use VisDrone2019-DET validation, a local COCO-style UAVDT conversion, and AI-TOD validation [1], [2], [3]. The available AI-TOD crop reconstruction has xView-derived provenance [15]. Vehicles are mapped as VisDrone {car, van, truck, bus}, local UAVDT {car, truck, bus}, and AI-TOD {vehicle}. Raw coverage is determined from image names in a prediction artifact *before* class filtering; an executed image with no retained vehicle prediction is consequently evaluated as an empty prediction. Table I gives the support denominator. DOTA-v2 quadrilateral-to-HBB scale evidence is retained as a structural supplementary check, not as a metric or rank result [4].

AI-TOD has incomplete raw prediction coverage and is used in the support layer only. The metric and rank layers use the full VisDrone coverage; local UAVDT is retained for supplementary common-coverage and dense-threshold controls. The primary VisDrone pool comprises five fixed configurations: two resolutions of one baseline checkpoint (B-640 and B-1280), independently trained Y11m and ASD configurations, and a fine-tuned Y11n-P2 configuration. Every configuration has 548/548 raw-image coverage.

TABLE I
SUPPORT DENOMINATOR AND RETAINED VEHICLE GROUND TRUTH. COVERAGE IS RAW-PREDICTION IMAGE COVERAGE. ρ_{24} AND $\rho_{.015}$ ARE RETAINED FRACTIONS AT THE TWO HEADLINE RULES.

Source	Coverage	Vehicle boxes	ρ_{24}	$\rho_{.015}$
VisDrone	548/548	17,040	73.3%	89.8%
Local UAVDT	305/305	11,054	95.1%	97.4%
AI-TOD	1,869/2,804	59,904	9.7%	81.0%

B. Support and perturbation contract

For a ground-truth box (w, h) in an image (W, H), define absolute and normalized support as

$$s_a = \max(w, h), \qquad s_n = \max\left(\frac{w}{W}, \frac{h}{H}\right). \tag{1}$$

For source s and rule t, $G_s(t)$ denotes valid mapped boxes retained by the rule, and $\rho_s(t) = |G_s(t)|/|G_s|$ is the retained-support fraction. We register $s_a \in \{16, 20, 24, 28, 32, 40, 48, 64\}$ pixels and $s_n \in \{.005, .0075, .010, .015, .020, .030\}$; 24 pixels and .015 are the headline interior values.

Figure 2(a) defines three primary policies. I retains all mapped valid boxes and enables source ignores. O removes below-support valid boxes and disables source-ignore neutralization. N uses the same filtered target as O, while enabling source ignores after valid-GT matching. Under all three policies, a prediction overlapping a removed valid box remains eligible as a false positive; the separate removed-as-ignore control is supplementary. For a fixed support rule and metric m, the fixed-rule policy band is

$$B_m^\pi(t) = \max_{\pi \in \{I,O,N\}} m(t, \pi) - \min_{\pi \in \{I,O,N\}} m(t, \pi). \tag{2}$$

We use global micro-F_1 at confidence .25 and IoU .25, with score-ordered, valid-GT-first greedy matching. Source-ignore neutralization uses the COCO crowd overlap convention: an otherwise unmatched prediction is suppressed when its intersection-over-prediction-area is at least .5 [16]. AP-cap and matching controls are reported separately in the supplementary material.

C. Common-coverage rank protocol

For two configurations a, b and a registered rule r, the pairwise margin is $\Delta_{ab}(r) = m_a(r) - m_b(r)$. We evaluate margins only on the common image set and treat exact ties as unstable. The headline comparison fixes policy N and IoU .25. A supplementary 14-threshold-by-eight-confidence sensitivity surface locates finite-grid decision boundaries; it does not extrapolate beyond the declared grid. For headline and observed-boundary cells, a paired bootstrap resamples the 548 images jointly for both configurations 10,000 times and reports the percentile 95% interval of Δ_{ab}.

III. RESULTS

A. Layer 1: Annotation-support mismatch

Figure 1 is the first link in the chain. A common 24-pixel threshold changes the retained target from nearly all local-UAVDT vehicles to fewer than one tenth of AI-TOD vehicles. The .015 normalized rule narrows this disparity, retaining 81.0–97.4% across the three sources at the headline value. The result is not that one support domain is universally preferable; it is that the target shift is measurable and should accompany the score that follows it.

The coverage record also decides which sources can progress to the later layers. AI-TOD's covered subset contains 16,900 vehicle boxes in 1,869 images (9.0 boxes/image), whereas its 935 noncovered images contain 43,004 boxes (46.0 boxes/image). We therefore retain AI-TOD for support diagnosis rather than a model comparison. The complete VisDrone and local-UAVDT records, by contrast, preserve images with empty retained predictions and their false-negative contribution.

B. Layer 2: Controlled support perturbation and metric uncertainty

The controlled perturbation makes the score response interpretable. At 24 pixels, F_1 policy bands range from .031 (Y11m) to .093 (B-1280); at .015, they range from .013 (Y11n-P2) to .035 (Y11m). Filtering can raise or lower a score because it changes the valid target and the treatment of unmatched predictions. Figure 2 therefore reports uncertainty of the evaluation contract, rather than attributing a policy-sensitive score to an unobserved annotation-error mechanism.

The O–N comparison isolates source-ignore handling at the same filtered target. At 24 pixels, enabling valid-first source-ignore neutralization moves F_1 by .008–.022 across the five configurations. The larger I–filtered separations in Fig. 2 show why the support rule and removed-object treatment must be stated together: a single endpoint alone cannot reveal which evaluative change produced it.

C. Layer 3: Rank robustness on common coverage

The headline comparison is made on exactly the same 548 images for all five configurations. At absolute 24 pixels, Y11m has $F_1 = .8181$ versus .7732 for ASD, giving $\Delta = .0449$ with a paired 95% interval [.0391,.0506]. At normalized .015, the corresponding values are .8741 and .8465, with $\Delta = .0277$ [.0234,.0321]. Y11m remains the winner and ASD the runner-up at both declared headline rules; the lower B-640/B-1280 order changes, which Fig. 3(a) makes visible rather than averaging away.

Supplementary sensitivity surfaces identify where a point ordering is not an established advantage. At the normalized .010/confidence-.30 boundary, Y11m–ASD is .00015 [$-$.00435, .00477]; at the absolute-64-pixel/confidence-.35 boundary, B-640–Y11m is .00095 [$-$.00492, .00677]. Table II records the four paired comparisons. The conclusion is therefore scoped to the declared headline rules: their Y11m–ASD margins are resolved on common coverage, whereas near-boundary ranks require an uncertainty qualifier.

This contrast separates score movement from a model conclusion. A fixed rule can move several configurations in the same direction while preserving their pairwise separation. Conversely, a small differential response can overturn a near tie. The sensitivity surface locates these finite-grid decision regions;

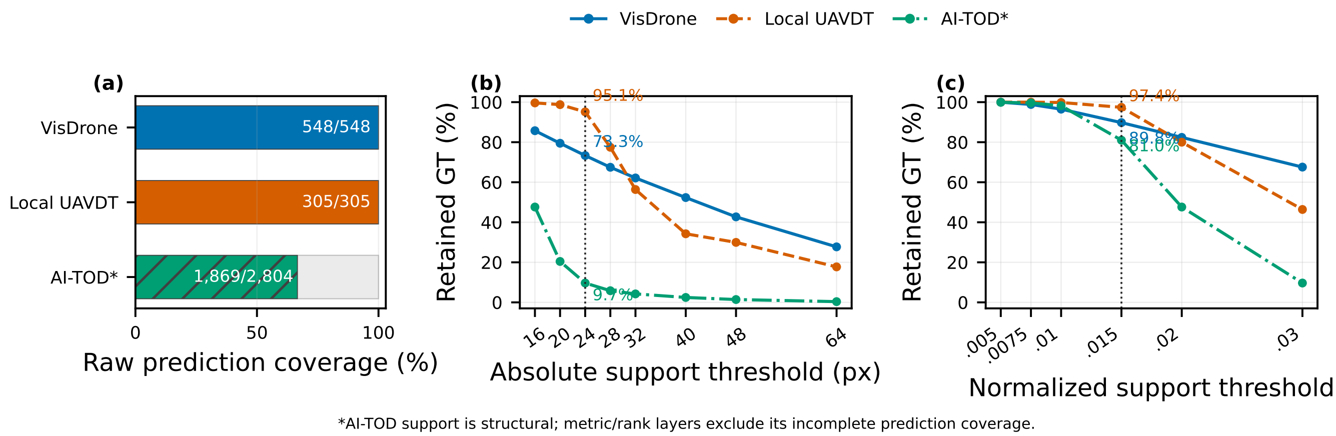

Fig. 1. Annotation-support mismatch. (a) Raw prediction coverage establishes the evaluation denominator before vehicle-class filtering. (b) A 24-pixel cutoff retains 73.3%, 95.1%, and 9.7% of vehicle ground truth in VisDrone, local UAVDT, and AI-TOD. (c) The normalized .015 rule retains 89.8%, 97.4%, and 81.0%. AI-TOD is displayed for support diagnosis only because its prediction coverage is incomplete.

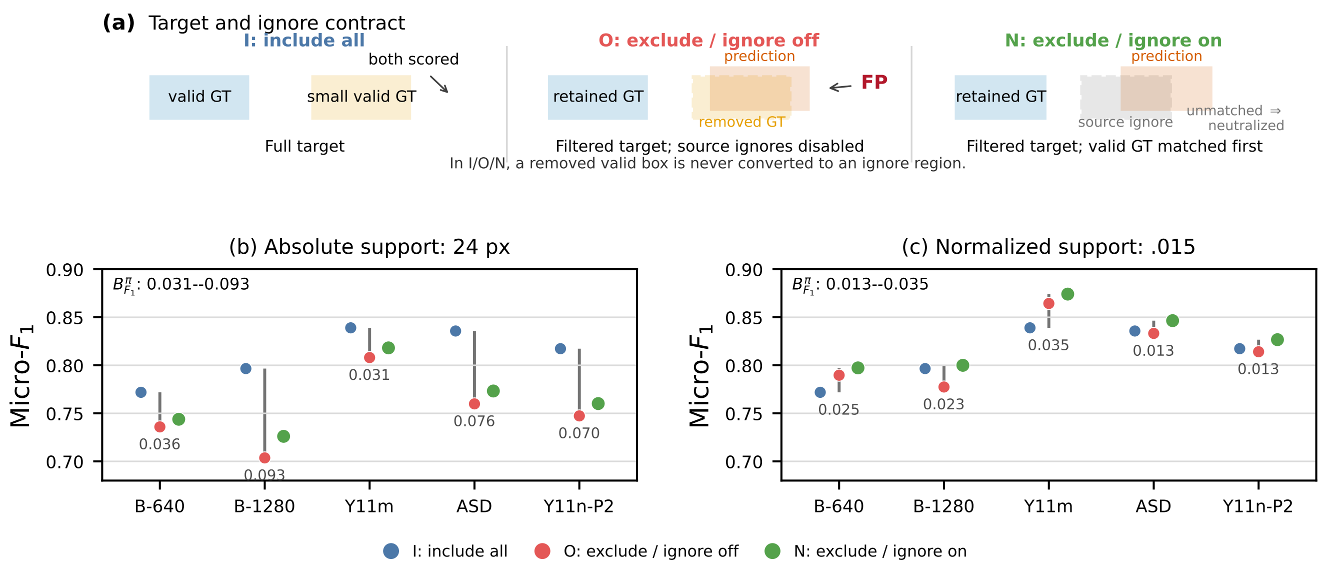

Fig. 2. Controlled support perturbation and metric uncertainty on the five-configuration VisDrone pool. (a) I, O, and N differ in target-set filtering and source-ignore treatment; removed valid boxes are not converted into ignore regions. (b),(c) Each vertical span is $B^{\pi}_{F_1}(t)$ for one configuration; colored points are I (blue), O (red), and N (green). The absolute-24-pixel bands span .031–.093, whereas normalized-.015 bands span .013–.035.

TABLE II
COMMON-COVERAGE PAIRED F_1 RECORDS (10,000 IMAGE-PAIRED REPLICATES). H = HEADLINE; B = OBSERVED FINITE-GRID BOUNDARY.

Cell	Pair	Δ	95% interval
H-a: 24 px, .25	Y11m–ASD	.0449	[.0391,.0506]
H-n: .015, .25	Y11m–ASD	.0277	[.0234,.0321]
B-n: .010, .30	Y11m–ASD	.00015	[−.00435,.00477]
B-a: 64 px, .35	B-640–Y11m	.00095	[−.00492,.00677]

the paired interval determines whether a selected pairwise advantage is resolved at the image level.

IV. Discussion and Reproducibility

The audit turns an evaluation question into a traceable sequence. First report how much annotation support remains; then expose how an explicit target/ignore perturbation moves the metric; finally state whether a frozen configuration-pool conclusion persists on common coverage and under paired uncertainty. This sequence separates target drift from metric sensitivity and from a model comparison.

For a compact comparison record, the three layers translate into three released objects: a source-level retained-support curve with the raw coverage denominator; a candidate-level table of policy endpoints and $B^{\pi}_{m}(t)$; and a pairwise common-coverage record containing the rule, point margin, resampling unit, and interval. These objects can accompany a conventional AP or F_1 table without altering a detector. They identify whether an

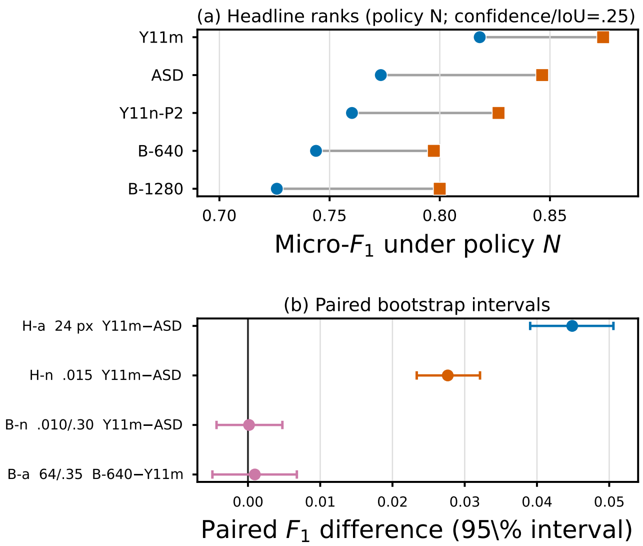

Fig. 3. Common-coverage rank robustness for five fixed VisDrone configurations. (a) Blue circles are absolute-24-pixel and orange squares normalized-.015 F_1 values under policy N; all configurations use the same 548 images. (b) H-a/H-n are the two headline comparisons and B-a/B-n are observed finite-grid boundary cells. Error bars are 95% image-paired bootstrap intervals from 10,000 resamples.

TABLE III
MINIMAL SUPPORT-CONDITIONED COMPARISON RECORD.

Layer	Released primitive and question resolved
Support	Raw image coverage, $\rho_s(t)$, and the support coordinate: *which valid target is scored?*
Perturbation	$I/O/N$ endpoints and $B_m^\pi(t)$: *how much does the metric move under the declared target/ignore contract?*
Rank	Common image set, pairwise Δ_{ab}, resampling unit, and interval: *is a pool-specific ordering resolved?*

apparent gain follows a changed target, a changed contract, or a reproducible candidate difference.

The record is deliberately ordered. Support can be measured from annotations before predictions are replayed; metric uncertainty then holds a prediction artifact fixed while the contract changes; rank evidence is finally meaningful only after the compared configurations share the same image universe. This ordering gives a reviewer or downstream user a direct path from a plotted score to the target and comparison on which it rests.

The supplementary material contains the complementary controls: full sensitivity surfaces, greedy–Hungarian matching checks, AP cap-contract checks, and structural DOTA-v2 evidence. These controls test evaluator boundaries without enlarging the central common-coverage rank claim. Code, configurations, derived evaluation records, figure-source tables, and checksums are available at https://github.com/Marchematics/aerial-evaluation-audit; licensed imagery and raw prediction artifacts remain referenced by manifest and hash.

V. CONCLUSION

Support-conditioned auditing connects annotation-support mismatch to a controlled metric perturbation and an uncertainty-qualified rank result. On the declared aerial artifacts, this chain exposes large cross-source target shifts, policy-dependent metrics, and a clear difference between robust headline margins and unresolved boundary rankings. The practical recommendation is to release the support denominator, perturbation contract, and common-coverage paired comparison with each aerial-detection result.

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